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ICLR

Dataless Weight Disentanglement in Task Arithmetic via Kronecker-Factored Approximate Curvature

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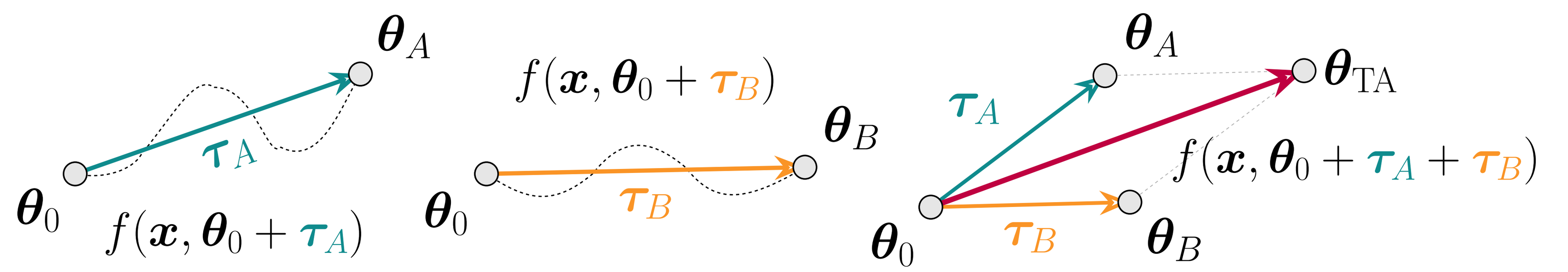
Background: Task Arithmetic

Idea. Fine-tune models, then combine weights (task vectors $\boldsymbol{\tau} = \boldsymbol{\theta} - \boldsymbol{\theta}_0$) through addition (***multitasking***) or subtraction (***unlearning***).

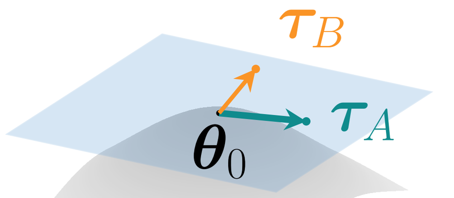


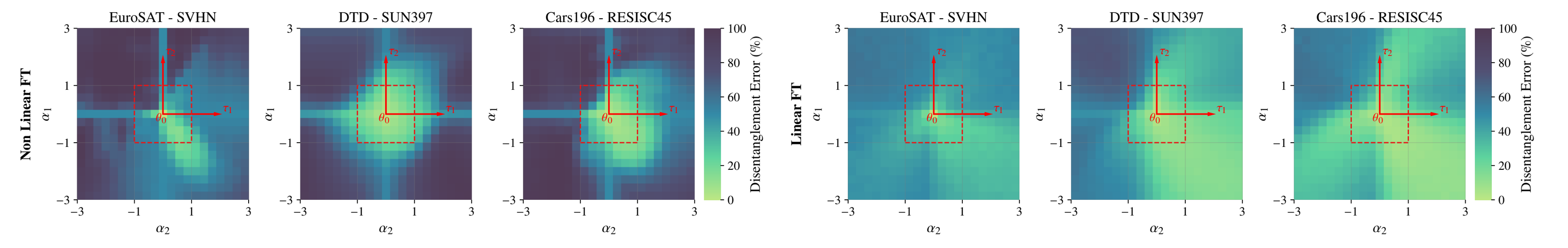
Weight Disentanglement. Data from distinct regions in input space affect separate, non-overlapping regions of the activation space.

$$f(\boldsymbol{x}, \boldsymbol{\theta}_0 + \boldsymbol{\tau}_A + \boldsymbol{\tau}_B) = f(\boldsymbol{x}, \boldsymbol{\theta}_0)\,\mathbb{I}_{\boldsymbol{x}\notin A\cup B} + f(\boldsymbol{x}, \boldsymbol{\theta}_0 + \boldsymbol{\tau}_A)\,\mathbb{I}_{\boldsymbol{x}\in A} + f(\boldsymbol{x}, \boldsymbol{\theta}_0 + \boldsymbol{\tau}_B)\,\mathbb{I}_{\boldsymbol{x}\in B}$$

Prior works. Fine-tune the **linearized model**.

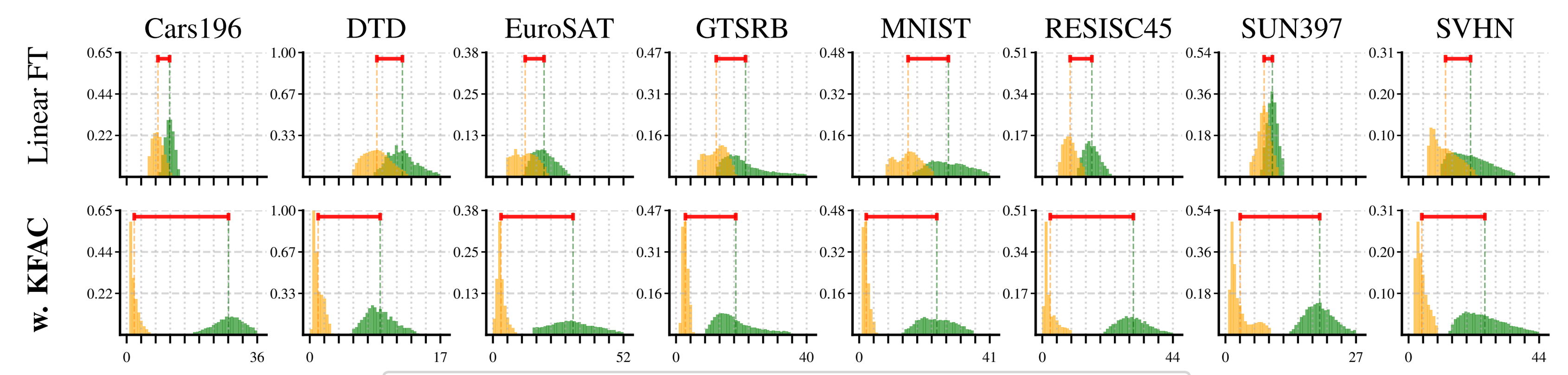
$$f_{\text{lin}}(\boldsymbol{x}, \boldsymbol{\theta}) = f(\boldsymbol{x}, \boldsymbol{\theta}_0) + \mathrm{J}_{\boldsymbol{\theta}} f(\boldsymbol{x}, \boldsymbol{\theta}_0)(\boldsymbol{\theta} - \boldsymbol{\theta}_0)$$





Reduced weight disentanglement error!

For effective task arithmetic, the first-order term $\mathrm{J}_{\boldsymbol{\theta}} f(\boldsymbol{x}, \boldsymbol{\theta}_0)(\boldsymbol{\theta}_t - \boldsymbol{\theta}_0)$ must vanish outside the training distribution; however, **this is not guaranteed**.



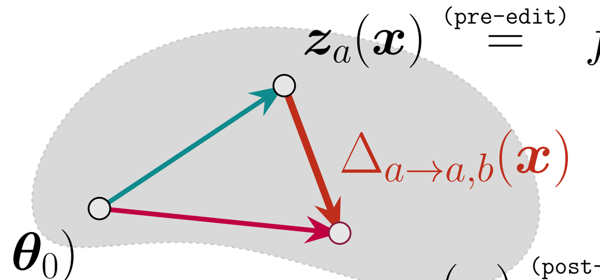
TL;DR: How to fine-tune for improved merging?

We fine-tune with K-FAC regularization to improve weight disentanglement, enabling effective Task Arithmetic & Model Merging without external data.

TAK: Task Arithmetic with KFAC regularization

STEP 1. Quantify the representation drift after merging.

$$\boldsymbol{z}_a(\boldsymbol{x}) \overset{\text{(pre-edit)}}{=} f_{\text{lin}}(\boldsymbol{x}, \boldsymbol{\theta}_0 + \alpha_a \boldsymbol{\tau}_a) \text{ s.t. } \boldsymbol{x} \in \mathcal{D}_a$$



$$\boxed{\|\boldsymbol{z}_{a,b}(\boldsymbol{x}) - \boldsymbol{z}_a(\boldsymbol{x})\|_2^2 = \alpha_b^2\, \|\mathrm{J}_{\boldsymbol{\theta}} f_{\text{lin}}(\boldsymbol{x}, \boldsymbol{\theta}_0)\, \boldsymbol{\tau}_b\|_2^2}$$

Representation drift

$$\boldsymbol{z}_{a,b}(\boldsymbol{x}) \overset{\text{(post-edit)}}{=} f_{\text{lin}}(\boldsymbol{x}, \boldsymbol{\theta}_0 + \alpha_a \boldsymbol{\tau}_a + \alpha_b \boldsymbol{\tau}_b)$$

STEP 2. Reformulate the drift over all examples $\boldsymbol{x} \in \mathcal{D}_a$.

$$\mathcal{L}^{\text{drift}}_{a\to a,b}(\boldsymbol{\tau}_b) = \alpha_b^2 \boldsymbol{\tau}_b^\top \boldsymbol{G}_a(\boldsymbol{\theta}_0) \boldsymbol{\tau}_b \quad \text{with} \quad \boldsymbol{G}_a(\boldsymbol{\theta}_0) = \frac{1}{|\mathcal{D}_a|} \sum_{\boldsymbol{x}\in\mathcal{D}_a} \mathrm{J}_{\boldsymbol{\theta}} f(\boldsymbol{x}, \boldsymbol{\theta}_0)^\top \mathrm{J}_{\boldsymbol{\theta}} f(\boldsymbol{x}, \boldsymbol{\theta}_0)$$

Precomputing $\boldsymbol{G}_a(\boldsymbol{\theta}_0)$ — No further data access required (dataless)

Issue — $\boldsymbol{G}_a$ is $|\boldsymbol{\theta}_0| \times |\boldsymbol{\theta}_0|$ (unfeasible)

STEP 3. Interpret $\boldsymbol{G}_t(\boldsymbol{\theta}_0)$ as a well-known matrix in optimization.

Under squared loss, $\boldsymbol{G}_t(\boldsymbol{\theta}_0)$ is the generalized Gauss-Newton curvature matrix.

STEP 4. Use KFAC to approximate $\boldsymbol{G}_a(\boldsymbol{\theta}_0)$.

$$\mathcal{L}^{\text{drift}}_{a\to a,b}(\boldsymbol{\tau}_b) = \alpha_b^2 \boldsymbol{\tau}_b^\top \boldsymbol{G}_a(\boldsymbol{\theta}_0) \boldsymbol{\tau}_b \overset{\text{KFAC}}{\approx} \alpha_b^2 \sum_{l=1}^{L} \boldsymbol{\tau}_b^{l\top} (\boldsymbol{B}_a^l \otimes \boldsymbol{A}_a^l) \boldsymbol{\tau}_b^l .$$

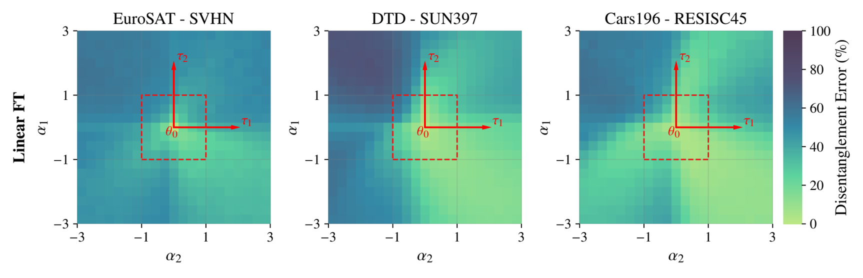
Final. Extend to ≥ 2 tasks.

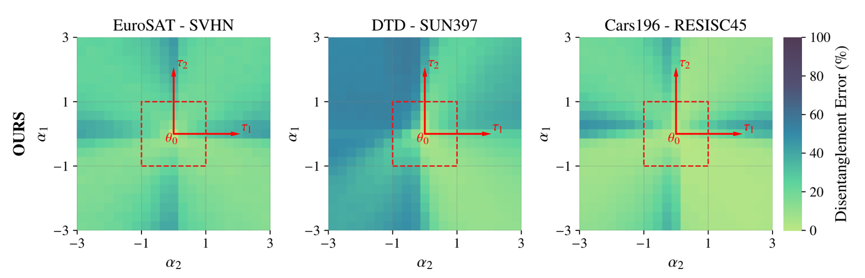
$$\boldsymbol{G}_{-t'}(\boldsymbol{\theta}_0^l) \overset{\text{KFAC}}{\approx} \sum_{t\neq t'} \lambda_t \boldsymbol{B}_t^l \otimes \boldsymbol{A}_t^l \overset{\text{merge}}{\approx} \Big(\sum_{t\neq t'} \boldsymbol{B}_t^l\Big) \otimes \Big(\sum_{t\neq t'} \lambda_t \boldsymbol{A}_t^l\Big).$$

Require: Network $f(\cdot, \boldsymbol{\theta}_0)$, target task $\mathcal{D}_{t'}$, other tasks $\{\mathcal{D}_t\}_{t\neq t'}$

Compute per-task GGNs $\{\boldsymbol{G}_{t\neq t'}\}$ (estimate via KFAC)
Merge curvature: $\boldsymbol{G}_{-t'} = \sum_{t\neq t'} \lambda_t \boldsymbol{G}_t$ (sum KFAC factors)
Linearize the net: $(f, \boldsymbol{\theta}_0) \to f_{\text{lin}}(\bullet, \boldsymbol{\tau}_{t'} - \boldsymbol{\theta}_0)$
while not converged **do**
    Draw a mini-batch $\mathcal{B} \sim \mathcal{D}_{t'}$
    Minimize $\mathcal{L}(\boldsymbol{\tau}_{t'}) = \mathcal{L}_{\text{CE}}(\mathcal{B}, \boldsymbol{\tau}_{t'}) + \beta \boldsymbol{\tau}_{t'}^\top \boldsymbol{G}_{-t'} \boldsymbol{\tau}_{t'}$
return task vector $\boldsymbol{\tau}_{t'}$

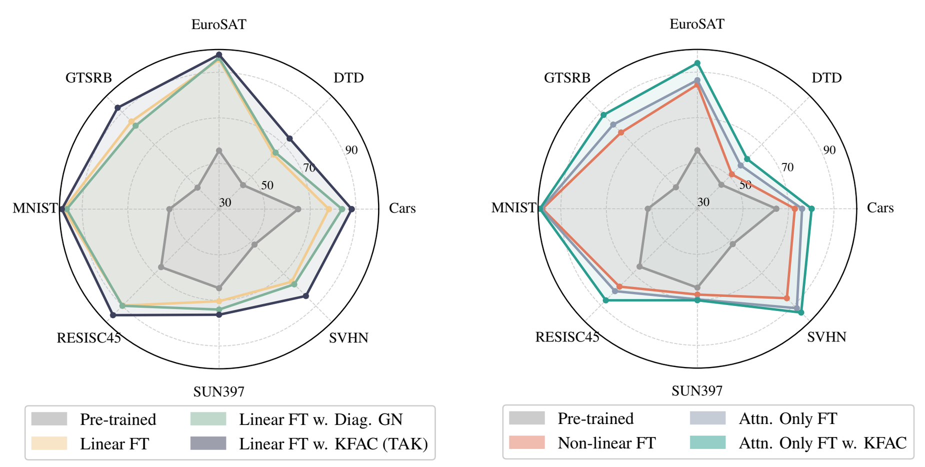
Experiments

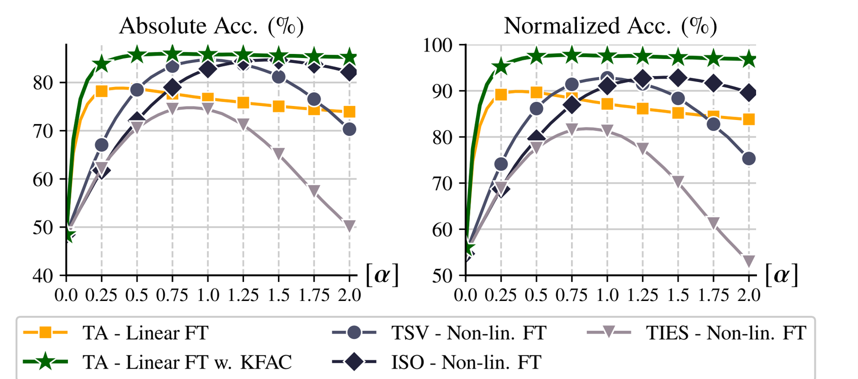




Better weight disentanglement!

Improved post-hoc merging!





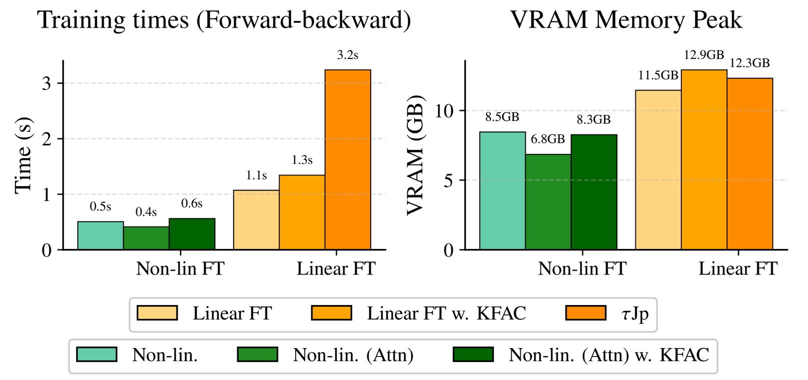
More robust to α scaling!

Method	Dataless	α	ViT-B/32 Abs.	ViT-B/32 Norm.	ViT-B/16 Abs.	ViT-B/16 Norm.	ViT-L/14 Abs.	ViT-L/14 Norm.
Pre-trained	–	–	48.4	–	55.4	–	65.0	–
Individual	–	–	90.9	–	92.4	–	93.8	–
Linear Fine-Tuning								
Linear FT	–	Best	78.8	89.9	82.0	90.9	88.0	94.8
τJp	×	Best	85.6	**98.2**	**88.6**	**98.7**	91.1	98.5
Diag. GGN	✓	Best	80.2	92.5	83.0	93.3	88.0	96.4
TAK, **Ours**	✓	Best	**86.0**	97.8	88.3	98.1	**91.6**	99.3
Non-Linear Fine-Tuning								
Non-linear FT	–	Best	73.5	80.4	77.0	82.9	84.5	89.7
Attn. Only FT	–	Best	78.2	86.3	80.4	87.1	88.2	93.8
TaLoS†	✓	Best	79.7	90.8	82.6	**92.4**	88.3	95.2
Attn. Only FT + TAK	✓	Best	**83.1**	**91.3**	**84.3**	91.0	**89.9**	**95.9**

SOTA on Task Addition & Negation.

Method	Dataless	ViT-B/32 Targ. ↓	ViT-B/32 Cont. ↑	ViT-B/16 Targ. ↓	ViT-B/16 Cont. ↑	ViT-L/14 Targ. ↓	ViT-L/14 Cont. ↑
Pre-trained	–	48.4	63.3	55.4	68.3	65.0	75.5
Non-linear FT	–	20.4	60.5	20.4	65.3	18.1	72.4
Linear FT	–	9.3	60.5	8.3	65.5	7.5	72.1
TaLoS†	✓	11.0	60.7	10.6	66.1	10.7	73.6
τJp	×	6.7	60.8	4.7	66.0	3.7	**73.0**
TAK, **Ours**	✓	**3.4**	**62.4**	**3.4**	**66.4**	**3.5**	72.6

KFAC runtime: 4 minutes.



Compressing KFAC factors